Engineering roles & disciplines

National 5 · Engineering contexts & challenges · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

☐ describe the roles of mechanical, electrical, electronic, civil, structural, chemical and environmental engineers

☐ tell the difference between an electrical and an electronic engineer

I DO — Worked example

Read how a full-mark answer is built

A new road is being built through farmland.

1 Name two types of engineer who would monitor the road construction and its impact on the land. (2 marks)

Points that earn marks from the marking instructions: one ✓ per point

✓ Civil engineer

✓ Environmental engineer

WE DO — Guided practice

Use the starters and word bank to help you

A new hospital is being designed.

2 Describe the role of an electrical engineer and an electronic engineer in the project. (2 marks)

Sentence starters

Electrical: …

Electronic: …

Word bank

standby · wiring · Power · supply · designs · electrical · lighting · distribution

Tip: 2 marks = 2 separate, developed points.

YOU DO — On your own

No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which engineer designs a bridge's foundations and road layout?

☐ Civil ☐ Electronic ☐ Chemical ☐ Electrical

b) Designing a gearbox for an e-bike is work for a…

☐ Structural engineer ☐ Civil engineer ☐ Chemical engineer ☐ Mechanical engineer

c) Monitoring how road building affects local wildlife is work for a…

☐ Mechanical engineer ☐ Electrical engineer ☐ Environmental engineer ☐ Chemical engineer

d) Who designs the control circuit and selects sensors?
☐ Civil ☐ Electronic ☐ Chemical ☐ Environmental

4 Key terms: write the letter of the matching description next to each term.

1. **Civil engineer**	____	**A** Large infrastructure: roads, bridges, dams, railways.
2. **Structural engineer**	____	**B** Machines, moving parts, drive systems and pneumatics.
3. **Mechanical engineer**	____	**C** Circuits, sensors, microcontrollers and control.
4. **Electrical engineer**	____	**D** Power generation, distribution, motors and wiring.
5. **Electronic engineer**	____	**E** Strength and stability of structures: forces and materials.

5 Explain what is meant by ‘Chemical engineer’. (2 marks)

6 Explain what is meant by ‘Environmental engineer’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?